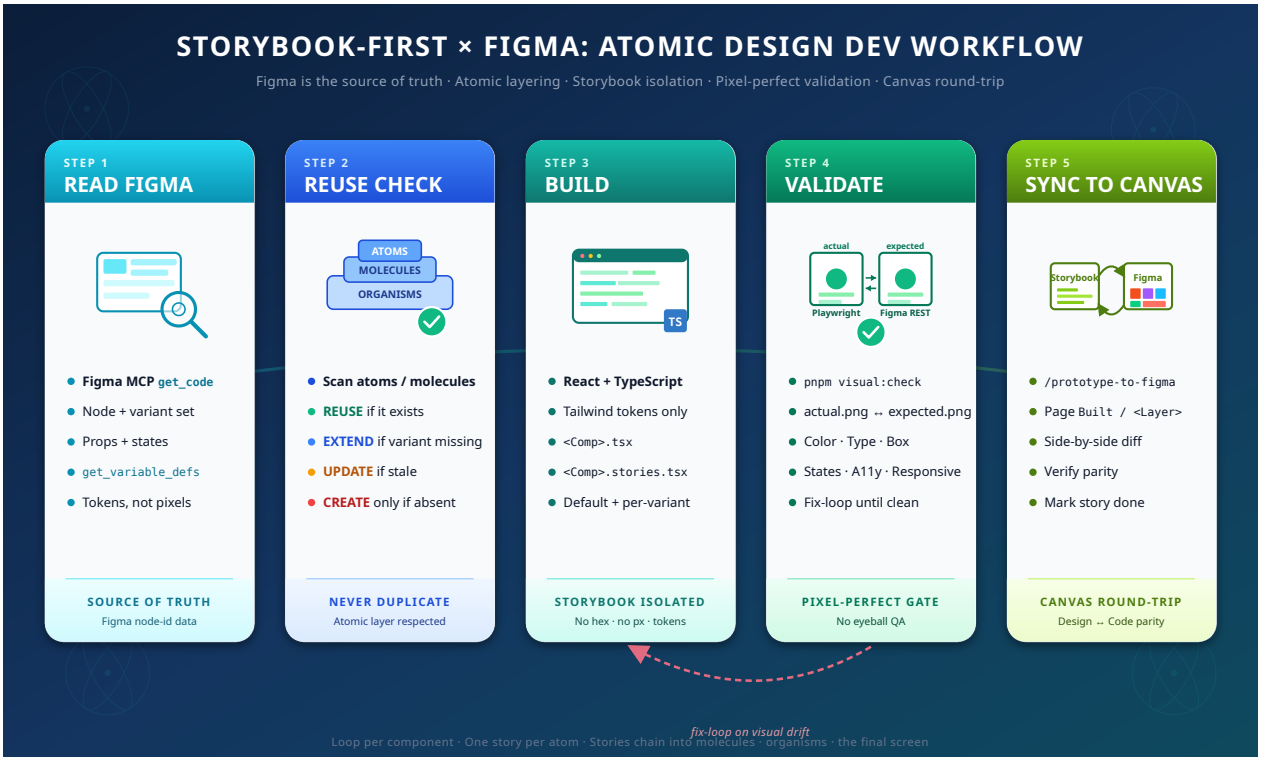
# Storybook-First with Figma

A bidirectional workflow for pixel-perfect atomic design

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## Problem

Two failure modes dominate UI work on real products.

**Visual approximation.** A button ends up with a hex that is *close enough* to the design. Spacing snaps to Tailwind's default scale instead of the exact Figma value. The font looks right but renders as Inter where Barlow Semi Condensed was specified. None of these get caught in code review — they surface when a stakeholder opens the page next to the Figma file and notices that nothing quite matches.

**Silent duplication.** Every developer builds their own button, their own badge, their own way of truncating text. Six months in, the project has three `<Button>` components and four different definitions of "primary blue."

Storybook-First with Figma is a workflow that closes both gaps. It runs inside a BMad-driven dev agent, bound to a five-step loop that ends with an automated screenshot diff against the Figma source.

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## The solution

Every UI story follows the same five-step per-component loop, prefaced by a Step 0 that runs once per story to handle decomposition and planning.

| # | Step | What happens |
|---|---|---|
| **0** | Refine the story | Resolve the Figma link; classify the target as atom / molecule / organism / screen; decompose recursively to find every lower-layer dependency; run a Reuse Check on each node (REUSE / EXTEND / UPDATE / OFF-LAYER / CREATE); produce a bottom-up build plan. |
| **1** | Read Figma | `get_code` + `get_variable_defs` via Figma MCP. Tokens come back named, never as raw hex. |
| **2** | Reuse Check | For each component the build plan touches: import if it exists, extend if a variant is missing, HALT if there's a design conflict or a brownfield off-layer match. |
| **3** | Build | React + TypeScript + Tailwind. Tokens only — never hardcoded hex/px. `<Component>.tsx` + `<Component>.stories.tsx` per Figma variant. |
| **4** | Validate | `pnpm visual:check` runs Playwright at 2x DPR on the Storybook story → `actual.png`; the Figma REST API renders the source node at 2x → `expected.png`. The agent reads both PNGs and walks the Visual Checklist (color, typography, box, states, a11y). Drift triggers a fix loop. |
| **5** | Sync to Canvas | `/prototype-to-figma` (a Figma MCP skill) pushes the built component to a `Built / <Layer>` page in the same Figma file. Designers verify parity side-by-side with the source frame. |

The 5-step loop is per-component. Step 0 decides which loops will run, and in what order — atoms first, then their consumers, finally the original target.

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## How it works

The module is a thin **execution discipline layer** on top of standard BMad. There is no fork, no patched skill — only one customization file and one helper script:

- `_bmad/custom/bmad-dev-story.toml` holds the policy. When the user invokes `/bmad-dev-story` in Claude Code (VS Code extension), BMad's customization resolver loads every entry of this file's `persistent_facts` array as foundational context for the run. That is where the **scope guard** (UI vs. NON-UI classification), the **preflight checklist** (Figma MCP reachable, `FIGMA_TOKEN` set, Playwright + Storybook present, npm scripts wired), the **Storybook lifecycle** (auto-start in background if not running), the 5-step loop, the visual-

check gate, the token-only rule, the layer enforcement, and Step 0 all live. The agent is bound by these rules regardless of who authored the story.
- `docs/workflows/storybook-first-with-figma.md` is the operating manual, loaded as a `file:` reference inside `persistent_facts`. It contains the per-step detail and the Visual Checklist.
- `scripts/visual-check.mjs` is a ~80-line script. Playwright screenshots the Storybook story at 2x DPR; the Figma REST API renders the source node at 2x. Both files land in `_visual-checks/<component>/<variant>/`. The agent (multimodal) reads both and walks the checklist. The script does its own Storybook reachability probe before launching Playwright, so a missing dev server fails with a clear message instead of a generic navigation timeout.

**Scope guard before everything.** The first thing the agent does is classify the story as UI or NON-UI. Backend, infra, schema, docs, and config stories skip the entire workflow and fall back to stock BMad — no Figma fetch, no visual-check, no Storybook requirement. The module only "activates" when it has something to enforce.

**Preflight before Step 0.** For UI stories, the agent runs a 6-item preflight check before any other work: Figma MCP reachable, `.env` has `FIGMA_TOKEN`, `scripts/visual-check.mjs` present, Playwright installed, `package.json` has `visual:check` script, Storybook set up. Each failure HALTs with the **exact** one-line remediation. Users never have to remember the manual setup steps — the agent surfaces them on demand.

**Adoption reality.** Most teams install this into projects that have been running for years. The first stories the team points it at will frequently target an organism or a screen with no atoms in `src/components/atoms/` yet. Step 0 handles this by decomposing the target recursively and building bottom-up in a single dev-story run — atoms first, then molecules, then organisms, then the target. Each component still passes through the 5-step loop and visual validation. The agent only HALTs in three cases: an **UPDATE conflict** (component exists at the right layer but differs from Figma), an **OFF-LAYER find** (component exists in the codebase but outside the atomic folder structure — common in brownfield), or an **explicit scope error** in the original story ("build all the molecules" — should be split via a discovery pattern).

**The Figma file matters.** The workflow's value is bounded by what's in the file. Variables for design tokens, atomic-design organization (frames named or grouped as Atoms / Molecules / Organisms / Screen), and Components with complete variant sets. Without them, tokens get derived from inline values, classification becomes guesswork, and Step 0 produces a noisier build plan.

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## Install and use

Open a terminal **inside the project root** you want to add the module to (the integrated terminal in VS Code works — files land in your working tree and show up in the Source Control panel

for review). Run:

```
bash <(curl -fsSL https://raw.githubusercontent.com/objectedge/story-book-first-with-figma/main/install.sh)
```

The script prompts only for your Figma file key, then auto-wires the project: drops 7 files into the current directory, copies `.env.example` to `.env` (with the file key prefilled, `FIGMA_TOKEN` left empty), adds `"visual:check": "node scripts/visual-check.mjs"` to `package.json` scripts, and installs Playwright + dotenv via the detected package manager ( `pnpm` / `yarn` / `npm` / `bun` ). Each auto-wiring step has an opt-out flag for teams that want full control. The Figma personal access token is **never** asked at install — the agent's preflight check guides the user to set it the first time a UI story runs.

After install, **one** thing to do:

- Open `.env` and paste a Figma personal access token in `FIGMA_TOKEN` (mint one at figma.com/settings → Personal access tokens).

Then invoke `/bmad-dev-story` with the path to a story your PM authored from a PRD (with a Figma link in its References section). The agent runs the preflight check — if anything else is missing (Figma MCP not yet approved via `/mcp` , Storybook not yet installed, dev server not running), it HALTs with the precise remediation instead of failing cryptically. Once preflight passes, it executes Step 0 (refinement + recursive build plan), confirms with you if more than three components will be created, then runs the 5-step loop per component, bottom-up. Every component lands in the correct layer folder, every Storybook story passes the Visual Checklist, every result is pushed back to Figma on a `Built / <Layer>` page.

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**Repository:** github.com/objectedge/story-book-first-with-figma · **License:** MIT